

Context Diagram and Data Flow Diagrams

Group 04 – LogiTeh

furlink

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Context Diagram

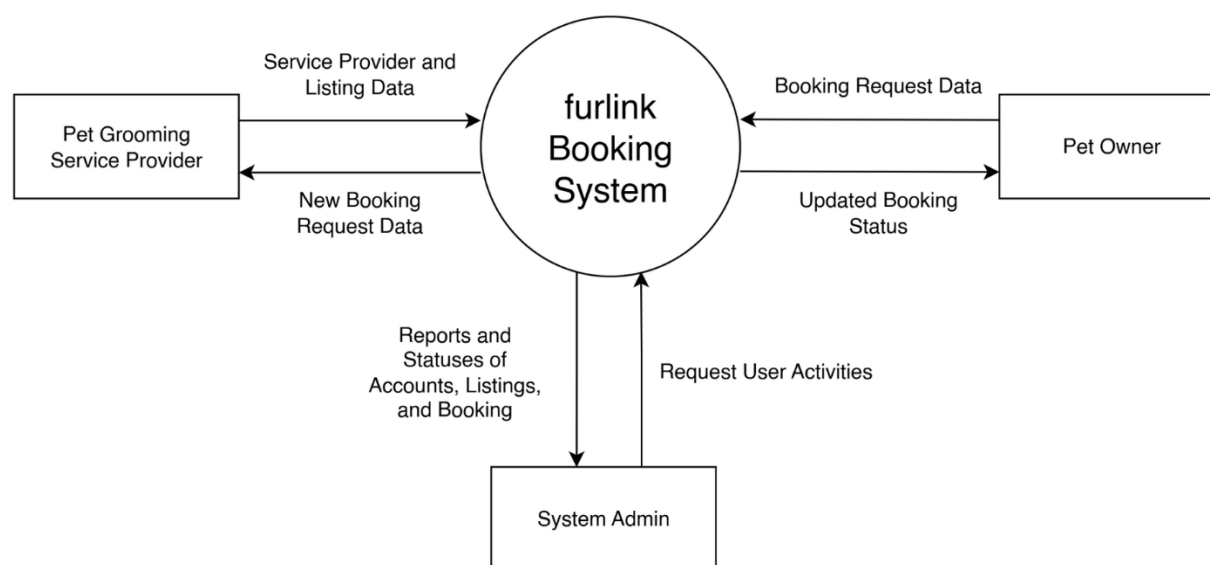


Figure 1 Context Diagram Level 0

The diagram above is for a booking system called *furlink* at high-level. It shows how the system interacts with external entities such as the service providers and the pet owners. Pet service providers can manage their listings by adding, viewing, updating, or deleting their services. They can also receive booking requests as initiated by pet owners. Meanwhile, pet owners enter their personal information along with details of their pets related to the grooming booking request. Once they request a booking appointment, pet owners receive confirmation of their scheduled appointment through the system.

On the other hand, a system administrator can request user activities. Once processed, the system will provide reports and statuses of accounts, listings, and bookings.

Level 1 Data Flow Diagram

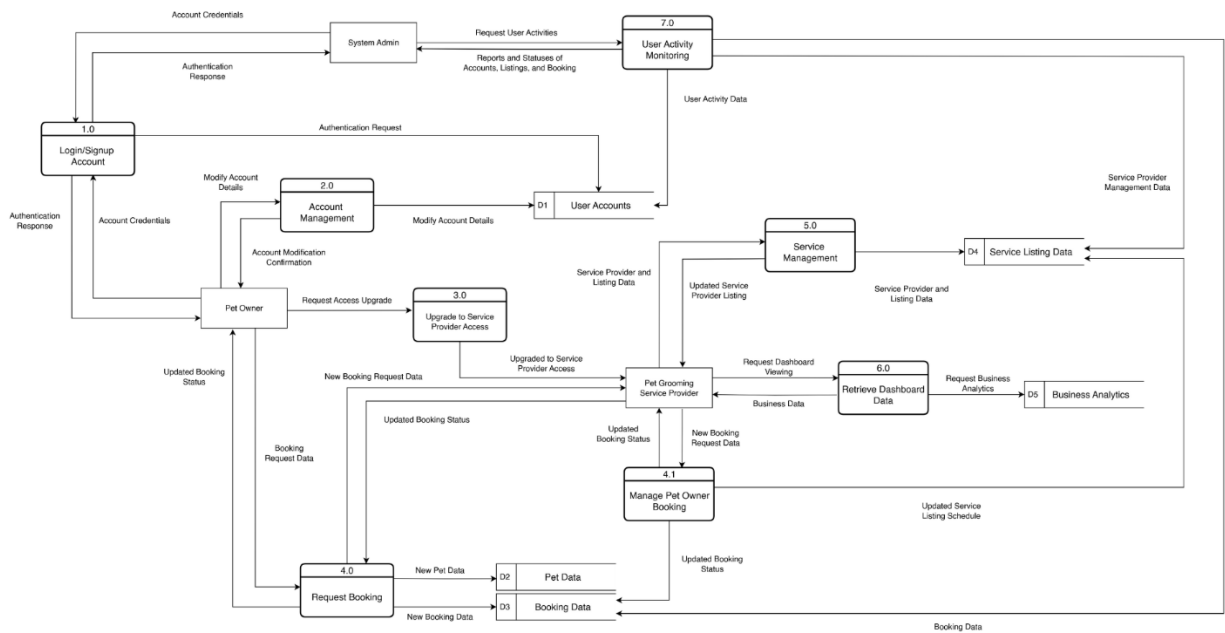


Figure 2 Level 1 Data Flow Diagram

The figure above is the Level 1 Data Flow Diagram which provides a more detailed view of furlink's system compared to the high-level Context Diagram. It shows the system's main processes, external entities, and data stores. A pet owner can Login or Signup through Process 1.0 and modify their account through Process 2.0, which interacts with the User Accounts data store. They can book services by sending a booking request with the pet and booking data which are stored in different data stores. To become a pet service provider, one must be logged in as a pet owner first and request to upgrade their access, so they are able to use service provider features. Once a pet owner has upgraded their access, they can now post and manage their services as service providers. A system admin can only manage user activities once they have logged in and have their credentials authenticated.

Level 2 Data Flow Diagrams

Login and Register

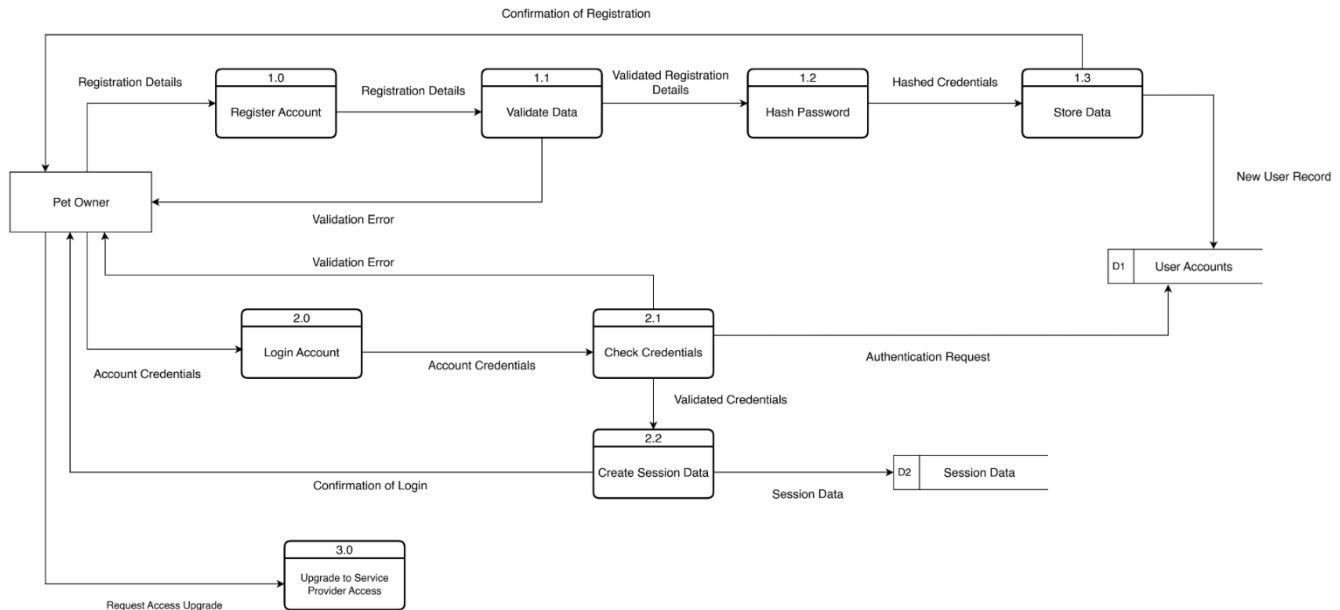


Figure 3 Level 2 DFD Pet Owner Login or Signup

The figure above is the Level 2 diagram which shows an extended and more detailed process of user login and registration. To login to the system, the user provides their email and password which the system validates. If their credentials match with the system, they are given access by providing them with a session ID. A session ID is created to maintain the user's authenticated state for all different interactions so that the system will be able to recognize the user after the login request. It also adds security because the credentials are not repeatedly sent after the authentication. However, if the credentials are found to be invalid, the system prompts the user that the credentials are incorrect.

On the other hand, if users want to create an account to access the system, they must first provide their email and password. They must confirm their password and input it twice, if the passwords do not match, it lets the user know that there was a validation error. They must also make sure that the email they input is a valid email. If their credentials are correct and matches, their password goes through a process of hashing them, so it is not stored as plain text on the data

store. The system then confirms to the user that their registration is successful once it is confirmed to be stored in the account data store. Lastly, process 3.0 is where the pet owner upgrades their account and be able to access features that are for pet service provider.

Pet Owner Account Management

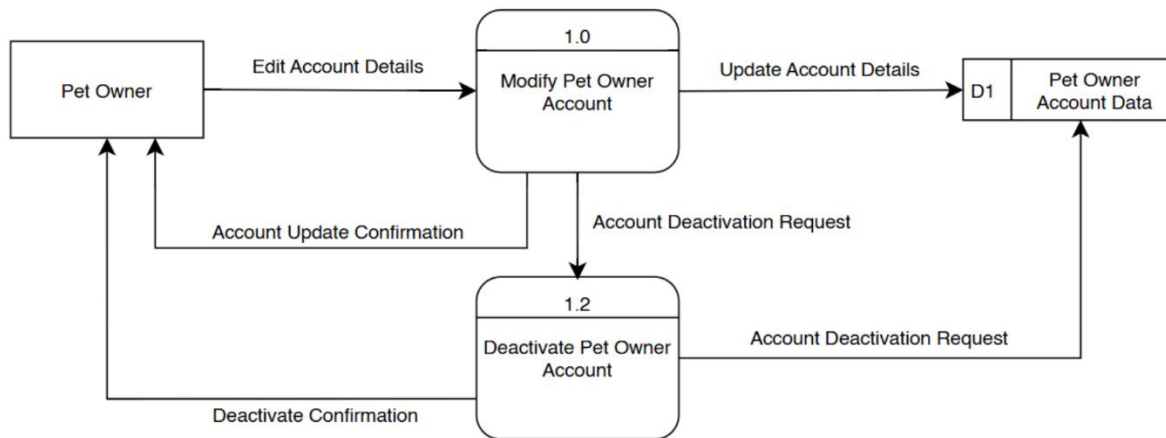


Figure 4 Level 2 DFD Pet Owner Account Management

The figure above shows a more detailed process of pet owner's account management, which allows pet owners to modify or deactivate their accounts. To successfully edit account details, the pet owner must modify the specific information they want, such as their name, password, etc. After this is done, the updated pet owner account information is stored in the pet owner account data, and the pet owner will receive a confirmation.

On the other hand, once the pet owner decides to deactivate their account, they send a deactivation request. Their account will be deactivated, and a confirmation will be sent back to the pet owner.

Booking

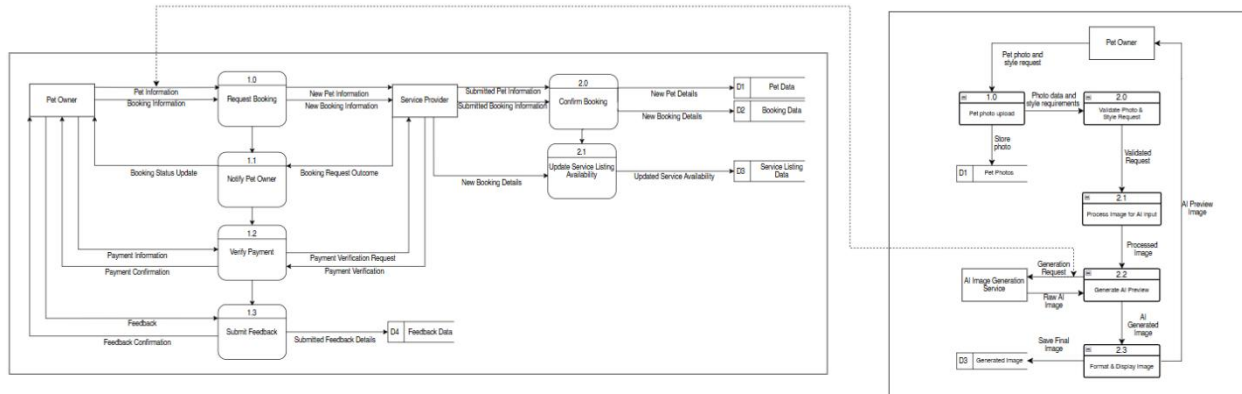


Figure 5 Level 2 DFD Booking & AI

The figure shows a detailed booking transaction between a pet owner and a pet service provider. The process begins with the pet owner submitting a booking request, which includes their pet and booking information. Additionally, as part of the booking process, the pet owner uploads a photo of their pet along with their style preferences.

This uploaded photo is stored in the Pet Photo Upload data store. The system then verifies the image quality and, if it's a valid request, it moves to the Process Image for AI Input step. This process prepares the pet photo for AI processing. Next, the Generate AI Preview process sends the prepared image to an external AI Image Generation Service and receives the generated AI image result. This result is then stored in the Generated Image data store before the final preview is displayed to the pet owner.

This new booking information is then sent to the service provider, who responds with either an approval or a denial. The pet owner is then notified of the status of their booking request.

Once a booking is approved, the pet owner makes a payment to the service provider outside of the system and sends the payment information to the provider. Upon receiving this information, the service provider verifies the payment and sends a verification response back to the pet owner.

After the payment is verified, the service provider confirms the booking information. The new pet and booking details are then stored in the Pet Data and Booking Data databases. At the same time, the service provider's service listing is updated to reflect the new booking. Once the payment verification is complete, the pet owner can submit feedback, which is then stored in the Feedback Data database, and the pet owner receives a confirmation.

Create, Read, Update, and Delete Service Provider

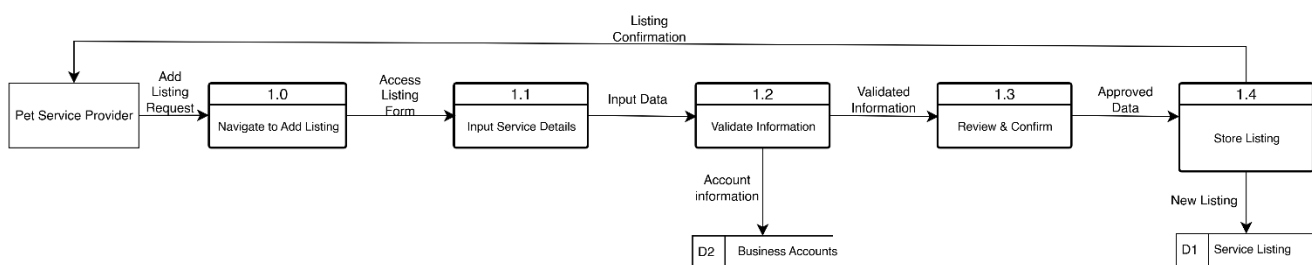


Figure 6 Level 2 DFD Create Service Listing

The figure above shows how pet service providers create their new service listing in furlink platform. It begins when pet service providers navigate to Add Listing, which provides access to the listing creation form. The Input Service Details collects the provider's service information and sends it to Validate Information then stores the information to the Business Accounts. The validated information then flows for Review and Confirm. The Approved Data then gets store in the Service Listing data store.

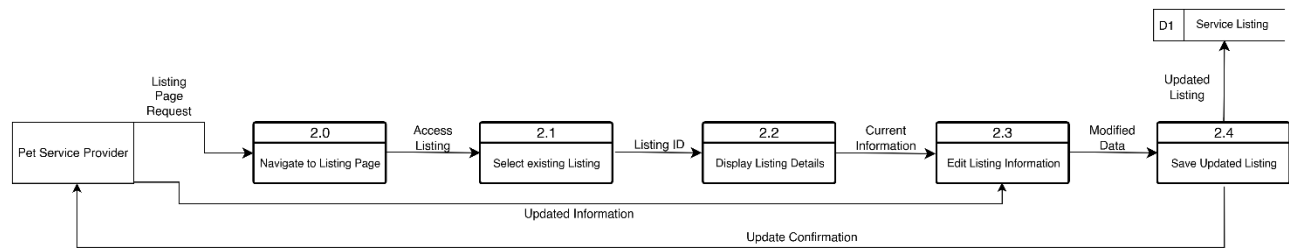


Figure 7 Level 2 DFD Update Service Listing

The figure above shows how pet service providers update the information in their service listing. It starts when the service provider Navigates to the listing page and access all the existing listing then selects an existing listing by providing the Listing ID. After the system uses the ID to Display Listing Details the service provider Edit the Listing Information and now has a Modified Data this is then used to Save Updated Listing and gets stored in the Service Listing data store. Finally, an Update Confirmation is sent back to the service provider.

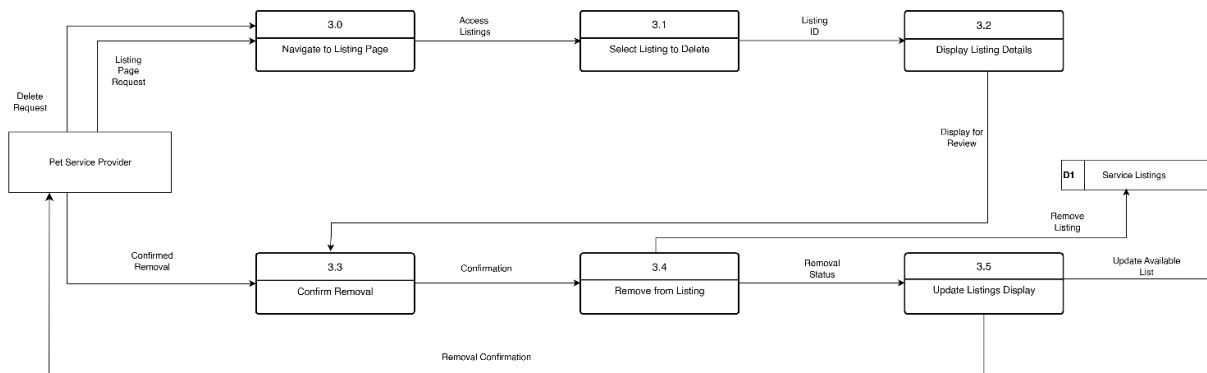


Figure 8 Level 2 DFD Delete Service Listing

The figure above shows how pet service providers remove their service listing. The process starts when the service provider navigates to the listing page and accesses the listings to select a listing to delete using the Listing ID. This will then display the listing details for the service provider to remove, they will then confirm removal. After confirmation, it will be removed from the listing and update the listing display and will now update the available list in the Service Listing data store.

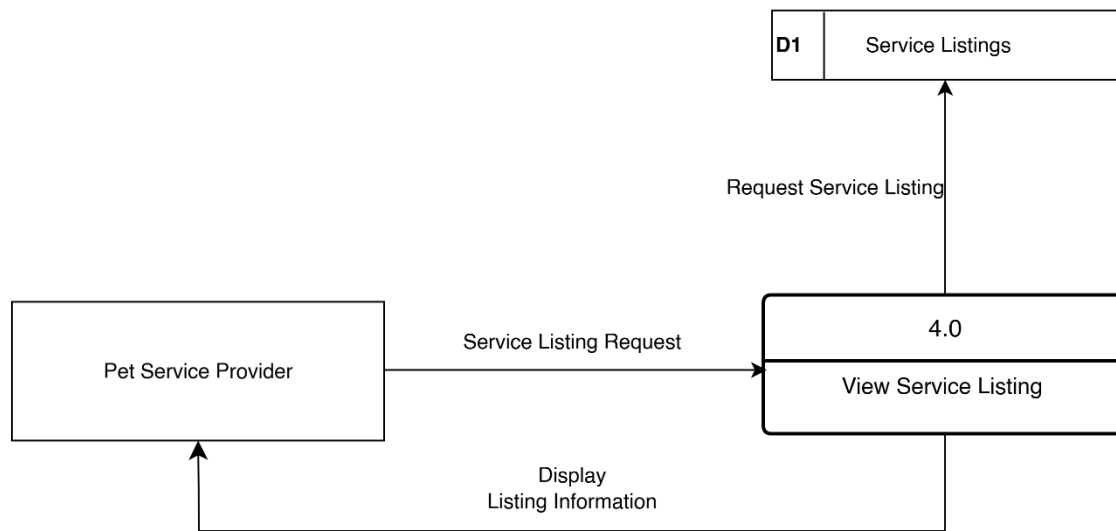


Figure 9 Level 2 DFD View Service Listing

The figure above shows how the pet service provider will view all their listing. It starts when the pet service provider sends a service listing request to view the service listing it will then send a request to the Service Listing data store then sends back to the pet service provider a display of the listing information.

System Administrator Manages User Activities

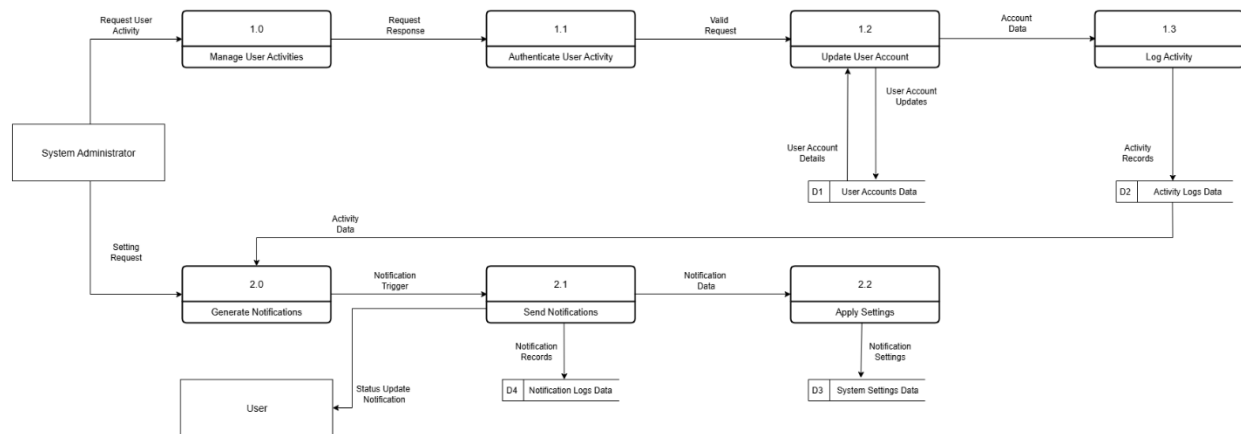


Figure 10 Level 2 DFD Administrator Manage User Activities

The figure shows the process flow for the system administrator managing user activities within furlink. The process will start once the system administrator submits a user activity request then this request will be forwarded to process 1.1 which will validate the system administrator's credentials and permissions.

After authentication, the valid request will proceed to process 1.2 where the user account will be updated with administrative actions. These actions include temporary suspensions, warnings, offboarding of service providers, or permanent suspension of pet owner accounts. The user account details will be retrieved from and updated in the User Accounts Data store. Following the account update, activity data flows from the account update process to process 2.0 which will create the notifications based on the actions done by the system administrator. These notifications will be processed by process 2.1 which will deliver the status update notifications to the user affected. Then, all notification records will be logged in the Notification Logs Data store.

Process 1.2 will send user account update information to process 1.3 which will store all the administrative actions in the Activity Logs Data Store. Additionally, the system administrator can modify the settings through the setting requests that flow to process 2.0 then to 2.2 which will update the System Settings Data store.

Dashboard Viewing

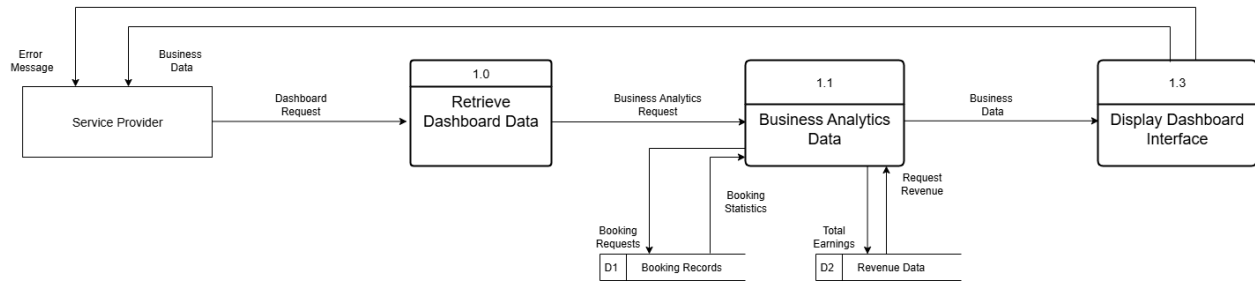


Figure 11 Level 2 DFD Dashboard Viewing

The figure shows the process of dashboard viewing, it begins when the service provider submits a dashboard request, to process 1.0 which will gather the necessary information and forwards a business analytics request to process 1.1. Then process 1.1 retrieves the booking statistics from data store 1 and revenue information from data store 2. This process generates business data that reflects the service provider's performance and financial information. Once the analytics process is complete, the business data flows from process 1.1 to process 1.2. This final process presents all the gathered business data in a user-friendly dashboard format which provides the service owner with a view of their business performance.